

Clinical Study Summary Report (CSSR)

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1. Administrative & Study Identification

Full Study Title: A Study to Investigate Safety, Tolerability, Pharmacokinetics, and Pharmacodynamics of AZD5492 in Adult Participants With Systemic Lupus Erythematosus or Idiopathic Inflammatory Myopathies or Rheumatoid Arthritis. **Short Title/Acronym:** TITAN **Protocol Number:** AstraZeneca Internal Protocol ID / TITAN **NCT Number:** NCT06916806 **Sponsor Name:** AstraZeneca (in collaboration with Fortrea) **Investigational Product:** AZD5492 (CD20xCD3 Trispecific T-cell engager / TriTE) **Phase of Development:** Phase 1 **Medical Monitor:** AstraZeneca Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective * **Primary Objective:** To assess the safety, tolerability, pharmacokinetics (PK) and pharmacodynamics (PD) of AZD5492 administered subcutaneously in adult participants with either Systemic Lupus Erythematosus (SLE), Idiopathic Inflammatory Myopathies (IIM), or Rheumatoid Arthritis (RA). * **Secondary Objectives:** To evaluate additional safety parameters, including the number of participants with abnormal electrocardiograms (ECGs) from Day 1 to Day 180, and to further characterize the PK/PD profile.

2.2 Background & Rationale To address the critical unmet medical need in patients with active, moderate-to-severe autoimmune diseases (SLE, IIM, RA) who are intolerant to or have an inadequate response to standard therapies. AZD5492 is a novel T-cell engaging bispecific antibody designed to reduce inflammation and control disease activity by targeting specific components of the immune system (CD20/CD8/CD3). This first-in-human study will establish the foundational safety and tolerability profile necessary for further clinical development.

3. Study Design & Methodology

3.1 Design Framework * **Study Type:** Interventional * **Control Type:** Single-arm * **Blinding:** Open-label * **Randomization:** Non-Randomized (Sequential Assignment)

3.2 Planned Enrollment & Duration * **Target \$N\$:** Phase 1 escalation cohorts (specific \$N\$ determined by Safety Review Committee depending on dose escalation steps). * **Number of Sites:** Multi-centre (including sites in the UK: London, Southampton,

Glasgow). * **Estimated Study Start:** April 2025. * **Estimated Primary Completion:** June 2027.

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Adults aged 18 to 70 years with a confirmed diagnosis of SLE (2019 EULAR/ACR criteria), IIM (2017 EULAR/ACR criteria), or RA. 2. For SLE: Positive for one or more of: anti-nuclear antibodies (titre $\geq 1:80$), anti-dsDNA, or anti-Sm at screening. Must have active, moderate-severe disease defined as a clinical SLEDAI-2K ≥ 4 . 3. Intolerance to, or inadequate response following at least 3 months of use to, ≥ 3 available treatments (e.g., corticosteroids, anti-malarial drugs, calcineurin inhibitors, methotrexate, belimumab, rituximab, etc.).

4.2 Exclusion Criteria 1. Significant history of or at risk of severe infections, HIV infection, positive COVID-19 PCR, or evidence of chronic/active hepatitis B or C. 2. Receipt of B-cell-depleting therapy, including CD19 or CD20 directed monoclonal antibodies (e.g., rituximab, ocrelizumab), < 3 months prior to Day 1. 3. Active severe, unstable or history of neuropsychiatric SLE, or significant CNS pathology.

5. Intervention & Treatment Plan

5.1 Investigational Regimen * **Dosage/Frequency:** * *Part 1 (Single Ascending Dose - SAD):* Single subcutaneous dose of AZD5492 on Day 1. * *Part 2 (Step-Up Dosing - SUD):* Subcutaneous administration at a priming dose on Day 1, followed by a target dose on Day 8 (two doses total). * **Route of Administration:** Subcutaneous (SC) injection. * **Duration of Treatment:** 1 to 2 doses within the first 8 days, followed by a minimum of 180 days of evaluation and up to 52 weeks (12 months) of safety follow-up.

5.2 Concomitant & Prohibited Therapy * **Permitted:** Stable background standard-of-care treatments not expressly prohibited by the protocol. * **Prohibited:** B-cell-depleting therapies (e.g., rituximab, ofatumumab, obinutuzumab) within 3 months prior to Day 1; live virus vaccines and strongly immunosuppressive rescue therapies unless required for patient safety.

6. Study Timeline & Visit Schedule

| Visit ID | Window (Days) | Key Activities/Procedures | | :---
| :--- | :--- | | **Screening** | Up to Day -1 | Informed Consent, Medical History, SLEDAI-2K/MMT-8 Scoring, Serology | | **Part 1 (SAD) Visits** | Days 1-4, 8, 15, 22, 30 | Single Dose (Day 1), PK/PD Labs, Safety Monitoring, Vitals | | **Part 2 (SUD) Visits** | Days 1-4, 8-11, 15, 22, 29, 43 | Priming Dose (Day 1), Target Dose

(Day 8), PK/PD Labs, Safety Monitoring | | **Observation** | Days 60, 90, 120, 150, 180 | Safety Labs, Efficacy Assessment, AE tracking | | **End of Study** | Up to Week 52 | Final Safety Assessments, Exit Interview |

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint * **Definition:** Safety evaluation of AZD5492, characterized by the number of participants with treatment-emergent adverse events (TEAEs), adverse events of special interest (AESIs), and serious adverse events (SAEs) from Day 1 to the end of the study (up to 52 weeks). * **Measurement Method:** Clinical assessments, investigator AE reporting, and laboratory evaluations.

7.2 Secondary & Exploratory Endpoints * Number and percentage of participants with abnormal ECG (from Day 1 up to Day 180). * Characterization of the Pharmacokinetics (PK) and Pharmacodynamics (PD) of AZD5492.

8. Safety & Adverse Event Reporting

- **Adverse Event (AE) Monitoring:** Continuous collection of AEs and AESIs via clinical evaluation and laboratory monitoring up to 52 weeks post-dose.
 - **Serious Adverse Events (SAEs):** Standard protocol timeline reporting (typically within 24 hours of investigator awareness).
 - **Data Monitoring Committee (DMC):** An internal Safety Review Committee (SRC) will evaluate emergent safety data from Part 1 to determine the priming and escalating target doses for Part 2.
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9. Statistical Analysis Plan (SAP) Summary

- **Analysis Populations:** Safety Analysis Set (all patients receiving ≥ 1 dose of AZD5492) and PK/PD Analysis Sets.
 - **Sample Size Justification:** The sample size is not based on formal statistical power calculations but rather on standard Phase 1 dose-escalation paradigms (SAD/SUD sequential design) to adequately assess safety and tolerability.
 - **Handling of Missing Data:** Standard descriptive statistics will be used. Missing safety data will generally not be imputed.
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10. Quality Assurance & Regulatory Compliance

- **GCP Compliance:** This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.
 - **Monitoring Plan:** Standard Phase 1 monitoring incorporating onsite visits, source data verification, and remote safety tracking.
 - **Audit Trail:** Comprehensive documentation using electronic Case Report Forms (eCRF) with full audit trails and data clarification mechanisms.
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11. Study Identifiers & Governance

- **Primary ID:** {{TITAN}}
 - **Registry Number:** {{NCT06916806}}
 - **Sponsor/Lead Organization:** {{AstraZeneca}}
 - **Study Title:** {{TITAN}}
 - **Official Regulatory Title:** {{A Study to Investigate Safety, Tolerability, Pharmacokinetics, and Pharmacodynamics of AZD5492 in Adult Participants With Systemic Lupus Erythematosus or Idiopathic Inflammatory Myopathies or Rheumatoid Arthritis}}
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12. Clinical Framework (Autoimmune Specific)

- **Primary Condition:** {{Systemic Lupus Erythematosus (SLE), Idiopathic Inflammatory Myopathies (IIM), Rheumatoid Arthritis (RA)}}
 - **Diagnosis Threshold:** {{Clinical SLEDAI-2K $\geq$ 4 for SLE; positive disease-specific autoantibodies for IIM}}
 - **Medical Methodology:** {{Target Induced T-cell Activating Nanobody (TITAN) Therapy / B-Cell Depletion}}
 - **Optimization Target:** {{Safety, Tolerability, and Immunomodulation}}
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13. Study Procedures & Methodology

- **Management Pathway:** {{Investigational T-cell engager pathway for refractory autoimmune disease}}
 - **Education Protocol (HCP):** {{Subcutaneous administration guidelines & Safety Review Committee reporting requirements}}
 - **Education Protocol (Patient):** {{Intensive Phase 1 visit adherence and AE self-reporting}}
 - **Quality Audit Mechanism:** {{Safety Review Committee (SRC) evaluation of dose-limiting toxicities prior to dose escalation}}
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14. Observation & Follow-Up Schedule

- **Total Duration:** {{Up to 52 Weeks (12 months)}}
- **Onsite Visit Cadence:** {{Screening, Days 1-4, 8-11, 15, 22, 29, 30, 43, 60, 90, 120, 150, 180}}
- **Remote Monitoring Frequency:** {{Continuous safety follow-up between scheduled visits}}
- **Checklist Review Interval:** {{As dictated by protocol up to Week 52}}

15. Sign-off & Approval

Role	Name	Signature	Date	---	---	---	---
Principal Investigator	[Insert Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Insert Name]	_____	[DD-MMM-YYYY]				
Lead Statistician	[Insert Name]	_____	[DD-MMM-YYYY]				